

Module 5 - Linear Transformations

S.1

Linear Transformation

Definition: Let $U(F)$ and $V(F)$ be two vector spaces. Then the function $T: U \rightarrow V$ is called a linear transformation from U into V if

$$T(a\alpha + b\beta) = aT(\alpha) + bT(\beta) \text{ for all } a, b \in F \text{ and } \alpha, \beta \in U.$$

Note: (1) If T is a linear transformation from U into itself (i.e., $T: U \rightarrow U$), then T is called a Linear operator on U .

(2) If $T: U \rightarrow F$ is a linear transformation, then T is called a linear functional on U .

(3) An n dimensional vector space V over a field F is denoted by $V_n(F)$.

Properties of Linear Transformation

Let $T: U(F) \rightarrow V(F)$ be a linear transformation from the vector space $U(F)$ to the vector space $V(F)$. Then

- (i) $T(\vec{0}) = \vec{0}$ where $\vec{0} \in U$ and $\vec{0} \in V$
- (ii) $T(-\alpha) = -T(\alpha)$ for all $\alpha \in U$
- (iii) $T(\alpha - \beta) = T(\alpha) - T(\beta)$ for all $\alpha, \beta \in U$
- (iv) $T(a_1\alpha_1 + a_2\alpha_2 + \dots + a_n\alpha_n) = a_1T(\alpha_1) + a_2T(\alpha_2) + \dots + a_nT(\alpha_n)$
for $a_1, a_2, \dots, a_n \in F$ and $\alpha_1, \alpha_2, \dots, \alpha_n \in U$.

Problems

① The mapping $T: V_3(\mathbb{R}) \rightarrow V_2(\mathbb{R})$ is defined by $T(x, y, z) = (x - y, x - z)$ for $(x, y, z) \in V_3$. Is T a linear transformation?

Sol: Let $\alpha = (x_1, y_1, z_1)$ and $\beta = (x_2, y_2, z_2)$ be any two vectors in V_3 .

For $a, b \in \mathbb{R}$,

$$\begin{aligned} T(a\alpha + b\beta) &= T(a(x_1, y_1, z_1) + b(x_2, y_2, z_2)) \\ &= T(ax_1 + bx_2, ay_1 + by_2, az_1 + bz_2) \\ &= ((ax_1 + bx_2) - (ay_1 + by_2), (ax_1 + bx_2) - (az_1 + bz_2)) \\ &= (a(x_1 - y_1) + b(x_2 - y_2), a(x_1 - z_1) + b(x_2 - z_2)) \\ &= a(x_1 - y_1, x_1 - z_1) + b(x_2 - y_2, x_2 - z_2) \\ &= aT(x_1, y_1, z_1) + bT(x_2, y_2, z_2) \\ &= aT(\alpha) + bT(\beta). \end{aligned}$$

Therefore, T is a linear transformation.

2. Is the mapping $T: \mathbb{R}^3(\mathbb{R}) \rightarrow \mathbb{R}^2(\mathbb{R})$ defined by $T(x, y, z) = (|x|, 0)$ a linear transformation?

Sol: Let $\alpha = (x_1, y_1, z_1)$, $\beta = (x_2, y_2, z_2) \in \mathbb{R}^3$.

For $a, b \in \mathbb{R}$, $T(a\alpha + b\beta) = T(a(x_1, y_1, z_1) + b(x_2, y_2, z_2))$

$$= T(ax_1 + bx_2, ay + by_2, az_1 + bz_2)$$

$$= (|ax_1 + bx_2|, 0)$$

$$\text{and } aT(\alpha) + bT(\beta) = aT(x_1, y_1, z_1) + bT(x_2, y_2, z_2)$$

$$= a(|x_1|, 0) + b(|x_2|, 0)$$

$$= (a|x_1| + b|x_2|, 0)$$

$$\text{Clearly } T(a\alpha + b\beta) \neq aT(\alpha) + bT(\beta)$$

$$(\text{since } |ax_1 + bx_2| \leq |a||x_1| + |b||x_2|)$$

Hence T is not a linear transformation.

3. Is $T: \mathbb{R}^3(\mathbb{R}) \rightarrow \mathbb{R}^3(\mathbb{R})$ defined by

$$T(x, y, z) = (x+1, y, z) \text{ a linear transformation}$$

Answer: not a L.T.

Note: Let $U(F)$ and $V(F)$ be two vector spaces

and $S = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ a basis of U . Let

$\{\beta_1, \beta_2, \dots, \beta_n\}$ be a set of vectors in V .

Then there exists a unique linear transformation

$T: U \rightarrow V$ such that $T(\alpha_i) = \beta_i$ for $i=1, 2, \dots, n$

4. Find a linear transformation $T: \mathbb{R}^2(\mathbb{R}) \rightarrow \mathbb{R}^2(\mathbb{R})$ is defined by $(T(1,1), T(1,0)) = (4,5)$ and $T(1,0) = (0,0)$.

Sol: Let $S = \{(2,3), (1,0)\}$. First we have to show that S is a basis of $\mathbb{R}^2$.

Let $(x,y) \in \mathbb{R}^2$. For $a, b \in \mathbb{R}$,

$$\begin{aligned}(x,y) &= a(2,3) + b(1,0) \\ &= (2a+b, 3a)\end{aligned}$$

$$\Rightarrow 2a+b=x \text{ and } 3a=y$$

$$\Rightarrow a = \frac{y}{3} \text{ and } b = \frac{3x-2y}{3}$$

$$\therefore (x,y) = \frac{y}{3}(2,3) + \left(\frac{3x-2y}{3}\right)(1,0)$$

and hence $L(S) = \mathbb{R}^2$. Clearly S is L.I.

Therefore, S is a basis of $\mathbb{R}^2$.

$$\text{Now, } T(x,y) = T\left(\frac{y}{3}(2,3) + \left(\frac{3x-2y}{3}\right)(1,0)\right)$$

$$= \frac{y}{3} T(2,3) + \frac{3x-2y}{3} T(1,0)$$

$$= \frac{y}{3} (4,5) + \frac{3x-2y}{3} (0,0)$$

$$= \left(\frac{4y}{3}, \frac{5y}{3}\right)$$

which is the required linear transformation

5. Find a linear transformation $T: \mathbb{R}^3(\mathbb{R}) \rightarrow \mathbb{R}(\mathbb{R})$ defined by $T(1, 1, 1) = 3$, $T(0, 1, -2) = 1$, $T(0, 0, 1) = -2$.

Answer: $T(x, y, z) = 8x - 3y - 2z$.

6. Find a linear transformation $T: V_2(\mathbb{R}) \rightarrow V_2(\mathbb{R})$ defined by $T(1, 2) = (3, 0)$ and $T(2, 1) = (1, 2)$.

Answer: $T(x, y) = \left(\frac{5y - x}{3}, \frac{4x - 2y}{3} \right)$.

Range space and Null space of a Linear Transformation

Definition: Let $U(F)$ and $V(F)$ be vector spaces and

$T: U \rightarrow V$ a linear transformation. Then the range of T is denoted by $R(T)$ and defined as

$$R(T) = \{ T(\alpha) \mid \alpha \in U \}$$

clearly $R(T)$ is a subspace of $V(F)$.

Definition: Let $U(F)$ and $V(F)$ be vector spaces

and $T: U \rightarrow V$ a linear transformation.

Then the null space of T is denoted by $N(T)$

and defined as $N(T) = \{ \alpha \in U \mid T(\alpha) = \hat{0} \}$,

$\hat{0}$ is the zero vector in V .

Clearly $N(T)$ is a subspace of $U(F)$.

The Null space $N(T)$ is also called as the Kernel of T .

Dimension of Range space and Null space

Definition: Let $T: U(F) \rightarrow V(F)$ be a linear transformation where U is a finite dimensional vector space. Then

(i) The dimension of the range space $R(T)$ is defined as rank of T and is denoted by $\rho(T)$. i.e., $\rho(T) = \dim R(T)$.

(ii) The dimension of the Null space $N(T)$ is defined as nullity of T and is denoted by $\nu(T)$. i.e., $\nu(T) = \dim N(T)$.

Note: $\boxed{\text{rank}(T) + \text{nullity}(T) = \dim U}$.

Problems:

① Let $T: \mathbb{R}^3(\mathbb{R}) \rightarrow \mathbb{R}^2(\mathbb{R})$ be a linear transformation defined by $T(x, y, z) = (x+y, y+z)$.

Then find the dimension of range space $R(T)$ and the null space $N(T)$.

Sol: Let $S = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ be the standard basis of $\mathbb{R}^3$.

Then $T(1, 0, 0) = (1, 0)$

$T(0, 1, 0) = (1, 1)$

$T(0, 0, 1) = (0, 1)$

Let $S_1 = \{(1, 0), (1, 1), (0, 1)\}$. Then $S_1 \subseteq R(T)$.

Now,
$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \xrightarrow{R_2 - R_1} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{bmatrix} \xrightarrow{R_3 - R_2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

Therefore, $\{(1, 0), (0, 1)\}$ forms a basis of $R(T)$.

Hence $\dim R(T) = 2$.

For $\alpha \in N(T)$, $T(\alpha) = \hat{0} \Rightarrow T(x, y, z) = (0, 0)$

$\Rightarrow (x+y, y+z) = (0, 0)$

$\Rightarrow x+y=0$

and $y+z=0$

Let $z = k$, then we get $y = -k$ and hence

$x = k, y = -k, z = k$.

$\therefore \left\{ \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} \right\}$ form a basis for $N(T)$ and hence

$\dim N(T) = \text{nullity}(T) = 1$

Therefore, $\dim R(T) + \dim N(T) = 2 + 1$

i.e., $\boxed{\text{rank}(T) + \text{nullity} = 3 = \dim \mathbb{R}^3}$

2. Let $T: \mathbb{R}^4(\mathbb{R}) \rightarrow \mathbb{R}^3(\mathbb{R})$ be a linear transformation defined by $T(a, b, c, d) = (a-b+4d, a+2c-d, a+b+3c-3d)$ for $a, b, c, d \in \mathbb{R}$, then find basis and dimension of $R(T)$ and $N(T)$. Answer: (i) $\{(1, 1, 1), (0, 1, 2)\}$ is a basis of $R(T)$ and $\dim R(T) = 2$, $\dim N(T) = \text{nullity}(T) = 2$. (ii) $\{(-2, -1, 1, 0), (1, 2, 0, 1)\}$ is a basis of $N(T)$.

Invertible Linear Transformation

Non-Singular Transformation :

Definition: A linear transformation $T: U(F) \rightarrow V(F)$ is said to be non-singular if the null space of T consists zero vector alone. [i.e., $N(T) = \{\hat{0}\}$, where $\hat{0}$ is the zero vector in $U(F)$.]

$$\text{i.e., } \alpha \in U \text{ and } T(\alpha) = \bar{0} \Rightarrow \alpha = \hat{0}.$$

Note: ① Let $U(F)$ and $V(F)$ be two finite dimensional vector spaces. The linear transformation $T: U \rightarrow V$ is one-one onto iff (if and only if) T is non-singular. So, $T: U \rightarrow V$ is invertible if and only if T is non-singular.

② A linear transformation $T: U(F) \rightarrow V(F)$ is said to be invertible if there exists a linear transformation $S: V(F) \rightarrow U(F)$ such that $T \circ S = S \circ T = I$, identity transformation. Here S is the inverse of T .

③ Let $U(F)$ and $V(F)$ be two vector spaces. Then the set $L(U, V)$ of all linear transformations from $U(F)$ into $V(F)$ is a vector space over a field F . This is called a vector space of linear transformations.

- ④ If $\dim U(F) = m$ and $\dim V(F) = n$, then $\dim(L(U, V)) = mn$.

Problems

- ① If a linear transformation $T: \mathbb{R}^3(\mathbb{R}) \rightarrow \mathbb{R}^3(\mathbb{R})$ defined by $T(x, y, z) = (2x, (4x - y), (2x + 3y - z))$ is invertible, then find T^{-1} .

Sol: Let $T(x, y, z) = (a, b, c)$

Then $T^{-1}(a, b, c) = (x, y, z)$
(since T is invertible)

Now, $T(x, y, z) = (a, b, c)$

$$\Rightarrow (2x, 4x - y, 2x + 3y - z) = (a, b, c)$$

$$\Rightarrow 2x = a, 4x - y = b, 2x + 3y - z = c$$

$$\Rightarrow x = \frac{a}{2}, y = 2a - b, z = 7a - 3b - c$$

$$\therefore T^{-1}(a, b, c) = \left(\frac{a}{2}, 2a - b, 7a - 3b - c\right)$$

- ② show that a linear transformation

$T: \mathbb{R}^3(\mathbb{R}) \rightarrow \mathbb{R}^3(\mathbb{R})$ defined by $T(x, y, z) = (x + y + z, y + z, z)$ is invertible and hence find T^{-1} .

Sol: we have T is invertible if T is non-singular

$$\text{For } \alpha = (x, y, z) \in \mathbb{R}^3, \quad T(\alpha) = \hat{0}$$

$$\Rightarrow T(x, y, z) = (0, 0, 0)$$

$$\Rightarrow (x+y+z, y+z, z) = (0, 0, 0)$$

$$\Rightarrow x+y+z=0, y+z=0, z=0$$

$$\Rightarrow x=0, y=0, z=0$$

$$\Rightarrow \alpha = \hat{0}$$

Therefore, $N(T) = \{\hat{0}\}$ and hence T is non-singular. Thus T is invertible.

$$\text{Let } T(x, y, z) = (a, b, c). \text{ Then } T^{-1}(a, b, c) = (x, y, z)$$

$$\text{Now, } T(x, y, z) = (a, b, c) \Rightarrow (x+y+z, y+z, z) = (a, b, c)$$

$$\Rightarrow x+y+z=a, y+z=b, z=c$$

$$\Rightarrow x=a-b, y=b-c, z=c$$

$$\therefore T^{-1}(a, b, c) = (a-b, b-c, c)$$

3. Show that a linear transformation $T: \mathbb{R}^3(\mathbb{R}) \rightarrow \mathbb{R}^3(\mathbb{R})$ defined by $T(x, y, z) = (3x, x-y, 2x+y+z)$ is invertible and hence find T^{-1} .

4. Show that a linear transformation $T: \mathbb{R}^2(\mathbb{R}) \rightarrow \mathbb{R}^2(\mathbb{R})$ defined by $T(x, y) = (x+y, x-y)$ is invertible and hence find T^{-1} .

5.2] Matrix of Linear Transformation

Let $U(F)$ and $V(F)$ be two finite dimensional vector spaces such that $\dim U = n$ and $\dim V = m$.
Let $T: U \rightarrow V$ be a linear transformation. Let $B_1 = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ be the ordered basis of U and $B_2 = \{\beta_1, \beta_2, \dots, \beta_m\}$ be the ordered basis of V .

For every $\alpha \in U$, $T(\alpha) \in V$ and $T(\alpha)$ can be expressed as a linear combination of the elements of the basis B_2 . So,

$$T(\alpha_1) = a_{11}\beta_1 + a_{21}\beta_2 + \dots + a_{m1}\beta_m$$

$$T(\alpha_2) = a_{12}\beta_1 + a_{22}\beta_2 + \dots + a_{m2}\beta_m$$

$\vdots$

$$T(\alpha_n) = a_{1n}\beta_1 + a_{2n}\beta_2 + \dots + a_{mn}\beta_m$$

writing the co-ordinates of $T(\alpha_1), T(\alpha_2), \dots, T(\alpha_n)$ successively as columns of a matrix, we get

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

This matrix represented as $[a_{ij}]_{m \times n}$ is called

the matrix of the linear transformation T with respect to the bases B_1 and B_2 . we can denote this by $[T: B_1, B_2]$ or $[T] = [a_{ij}]_{m \times n}$.

Here $\begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}, \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}, \dots, \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$ are

co-ordinate vectors of $T(\alpha_1), T(\alpha_2), \dots, T(\alpha_n)$ with respect to a basis B_2 respectively.

$$\text{i.e., } [T(\alpha_1)]_{B_2} = \begin{bmatrix} a_{11} \\ a_{21} \\ \vdots \\ a_{m1} \end{bmatrix}, [T(\alpha_2)]_{B_2} = \begin{bmatrix} a_{12} \\ a_{22} \\ \vdots \\ a_{m2} \end{bmatrix}, \dots, [T(\alpha_n)]_{B_2} = \begin{bmatrix} a_{1n} \\ a_{2n} \\ \vdots \\ a_{mn} \end{bmatrix}$$

Note: (i) Let $T: V(F) \rightarrow V(F)$ be a linear operator such that $\dim V = n$. If $B_1 = B_2 = B$ (say), then the above said matrix is called the matrix of T relative to the ordered basis B of V . It is denoted by $[T: B]$ or $[T]_B$.

(ii) Let $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ be a basis of an n -dimensional vector space $V(F)$. Then for any $\alpha \in V$, $\alpha = a_1 \alpha_1 + a_2 \alpha_2 + \dots + a_n \alpha_n$, where $a_1, a_2, \dots, a_n \in F$.

Therefore, the co-ordinate vector of α relative to the basis B is $[\alpha]_B = \begin{bmatrix} a_1 \\ a_2 \\ \vdots \\ a_n \end{bmatrix}$.

Problems

1. Let $T: \mathbb{R}^2(\mathbb{R}) \rightarrow \mathbb{R}^3(\mathbb{R})$ be a linear transformation defined by $T(x, y) = (x+y, 2x-y, 7y)$. Find the matrix of linear transformation $[T: B_1, B_2]$ where B_1 is the standard basis of $\mathbb{R}^2$ and B_2 is the standard basis of $\mathbb{R}^3$.

Sol: we have $B_1 = \{(1, 0), (0, 1)\}$ is the standard basis of $\mathbb{R}^2$ and $B_2 = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ is the standard basis of $\mathbb{R}^3$.

$$\text{Now, } T(1, 0) = (1, 2, 0) \\ = 1(1, 0, 0) + 2(0, 1, 0) + 0(0, 0, 1)$$

$$T(0, 1) = (1, -1, 7) \\ = 1(1, 0, 0) - 1(0, 1, 0) + 7(0, 0, 1)$$

$$\text{Therefore, } [T: B_1, B_2] = \begin{bmatrix} 1 & 1 \\ 2 & -1 \\ 0 & 7 \end{bmatrix}.$$

2. Let $T: \mathbb{R}^3(\mathbb{R}) \rightarrow \mathbb{R}^2(\mathbb{R})$ be a linear transformation defined by $T(x, y, z) = (3x+2y-4z, x-5y+3z)$. Find the matrix of T relative to the bases

$$B_1 = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\} \text{ and } B_2 = \{(1, 3), (1, 5)\}$$

Sol: let $(a, b) \in \mathbb{R}^2$ be such that

$$(a, b) = p(1, 3) + q(2, 5)$$

$$\text{then } (a, b) = (p+2q, 3p+5q)$$

$$\Rightarrow a = p+2q, \quad b = 3p+5q$$

$$\Rightarrow p = -5a+2b, \quad q = 3a-b$$

$$\therefore (a, b) = (-5a + 2b)(1, 3) + (3a - b)(2, 5).$$

$$\begin{aligned}\text{Now, } T(1, 1, 1) &= (1, -1) \\ &= -7(1, 3) + 4(2, 5)\end{aligned}$$

$$\begin{aligned}T(1, 1, 0) &= (5, -4) \\ &= -33(1, 3) + 19(2, 5)\end{aligned}$$

$$\begin{aligned}T(1, 0, 0) &= (3, 1) \\ &= -13(1, 3) + 8(2, 5).\end{aligned}$$

Therefore, the matrix of T relative to B_1 and B_2 is

$$[T: B_1, B_2] = \begin{bmatrix} -7 & -33 & -13 \\ 4 & 19 & 8 \end{bmatrix}$$

* Suppose $\alpha = (1, 1, 1)$, then $[T(\alpha)]_{B_2} = \begin{bmatrix} -7 \\ 4 \end{bmatrix}$ is the co-ordinate vector of $T(\alpha)$ relative to B_2 .

3. Let $T: \mathbb{R}^2(\mathbb{R}) \rightarrow \mathbb{R}^2(\mathbb{R})$ be a linear operator defined by $T(x, y) = (4x - 2y, 2x + y)$.

Find the matrix of T w.r.t. the basis $B = \{(1, 1), (-1, 0)\}$.

Sol: let $(a, b) \in \mathbb{R}^2$ be such that

$$(a, b) = p(1, 1) + q(-1, 0)$$

then $(a, b) = \cancel{(p-q)} + p(p-q, p)$

$$\Rightarrow a = p - q, \quad b = p$$

$$\Rightarrow p = b, \quad q = b - a$$

$$\therefore (a, b) = b(1, 1) + (b - a)(-1, 0)$$

Now, $T(1, 1) = (2, 3)$

$$= 3(1, 1) + 1(1, 0)$$

$$T(-1, 0) = (-4, -2)$$

$$= -2(1, 1) + 2(-1, 0)$$

Therefore, the matrix of T w.r.t. the basis

$$B = \{(1, 1), (-1, 0)\} \text{ is}$$

$$[T: B] = [T]_B = \begin{bmatrix} 3 & -2 \\ 1 & 2 \end{bmatrix}$$

* Here $\begin{bmatrix} 3 \\ 1 \end{bmatrix}, \begin{bmatrix} -2 \\ 2 \end{bmatrix}$ are the co-ordinate vectors

5.3 Change of Basis

Let $V(F)$ be an n -dimensional vector space.

Let $B = \{\alpha_1, \alpha_2, \dots, \alpha_n\}$ and $B' = \{\beta_1, \beta_2, \dots, \beta_n\}$ be two arbitrary bases for V . Let us suppose

$$\begin{aligned}\beta_1 &= a_{11}\alpha_1 + a_{21}\alpha_2 + \dots + a_{n1}\alpha_n \\ \beta_2 &= a_{12}\alpha_1 + a_{22}\alpha_2 + \dots + a_{n2}\alpha_n \\ &\vdots \\ \beta_n &= a_{1n}\alpha_1 + a_{2n}\alpha_2 + \dots + a_{nn}\alpha_n\end{aligned}$$

Then the matrix of transformation from B to B'

is
$$P = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}.$$

Here, the matrix P is called the transition matrix from B to B' .

(or, the change of basis matrix)

Note: Since $B' = \{\beta_1, \beta_2, \dots, \beta_n\}$ is L.I.,

the matrix P is invertible and P^{-1} is called the transition matrix from B' to B .

Examples

1. Let $B = \{(1, 0), (0, 1)\}$ and $B' = \{(1, 3), (2, 5)\}$ be the bases of $\mathbb{R}^2(\mathbb{R})$. Find the transition matrix from B to B' . Also find the transition matrix from B' to B .

Sol: (i) clearly, $(1, 3) = 1(1, 0) + 3(0, 1)$
 $(2, 5) = 2(1, 0) + 5(0, 1)$

Therefore, the transition matrix from B to B' is $\begin{bmatrix} 1 & 2 \\ 3 & 5 \end{bmatrix}$.

(ii) For any $(x, y) \in \mathbb{R}^2$,

$$(x, y) = a(1, 3) + b(2, 5)$$

$$\Rightarrow x = a + 2b, \quad y = 3a + 5b$$

$$\Rightarrow a = 2y - 5x; \quad b = 3x - y.$$

$$\therefore (x, y) = (2y - 5x)(1, 3) + (3x - y)(2, 5)$$

Now,

$$(1, 0) = -5(1, 3) + 3(2, 5)$$

$$(0, 1) = (+2)(1, 3) - 1(2, 5)$$

Therefore, the transition matrix from B' to B is $\begin{bmatrix} -5 & 2 \\ 3 & -1 \end{bmatrix}$.

② Find the transition matrix from $B = \{(1, 2), (3, 4)\}$ to $B' = \{(-1, 3), (4, 2)\}$, where B and B' are bases of a vector space $\mathbb{R}^2(\mathbb{R})$.

③ Find the transition matrix from $B = \{(9, 2), (4, -3)\}$ to $B' = \{(2, 1), (-3, 1)\}$, where B and B' are the bases of $\mathbb{R}^2(\mathbb{R})$.

④ Find the transition matrix from $B = \{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ to $B' = \{(1, 1, 0), (0, 1, 2), (1, 1, 1)\}$, where B and B' are the bases of $\mathbb{R}^3(\mathbb{R})$.

Similarity of Matrices

Definition: Let A and B be square matrices of order n over a field F . Then B is said to be similar to A if there exists an invertible matrix C of order n with elements in F such that $B = C^{-1} A C$.



Practice all the questions

1. Is a mapping $T: \mathbb{R}^3(\mathbb{R}) \rightarrow \mathbb{R}^3(\mathbb{R})$ defined by $T(x, y, z) = (x+y-z, 2x-y+z, -x+y+2z)$ a linear transformation?

Answer: Yes

2. Which of the following functions are linear transformations from $\mathbb{R}^2(\mathbb{R})$ into $\mathbb{R}^2(\mathbb{R})$.

(i) $T(x, y) = (x^2, y)$ (ii) $T(x, y) = (x, |y|)$ (iii) $T(x, y) = (x-y, 0)$

Answer: (i) not a linear transformation (ii) not a linear transformation (iii) a linear transformation

3. Find the linear transformation $T: \mathbb{R}^2(\mathbb{R}) \rightarrow \mathbb{R}^2(\mathbb{R})$ such that $T(2, 3) = (4, 5)$ and $T(1, 0) = (0, 0)$.

Answer: $T(x, y) = \left(\frac{4y}{3}, \frac{5y}{3} \right)$.

4. Let $T: \mathbb{R}^3(\mathbb{R}) \rightarrow \mathbb{R}^3(\mathbb{R})$ be a linear transformation defined by

$T(x, y, z) = (2x, 4x-y, 2x+3y-z)$ for $x, y, z \in \mathbb{R}$. Show that T is invertible and find T^{-1} .

Answer: $T^{-1}(a, b, c) = \left(\frac{a}{2}, 2a-b, 7a-3b-c \right)$.

5. Let $T: \mathbb{R}^2(\mathbb{R}) \rightarrow \mathbb{R}^3(\mathbb{R})$ be a linear transformation defined by $T(x, y) = (x+y, x-y, y)$. Find the rank and nullity of T .

Answer: rank of $T = 2$ and nullity of $T = 0$.

6. Let $T: \mathbb{R}^3(\mathbb{R}) \rightarrow \mathbb{R}^3(\mathbb{R})$ be a linear transformation defined by

$T(x, y, z) = (y+z, x-z, -x-y)$. Find the matrix of the linear transformation T with respect to the basis $B = \{(0, 1, -1), (1, 0, -1), (1, -1, 0)\}$.

Answer: $[T]_B = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

7. Let $T: \mathbb{R}^3(\mathbb{R}) \rightarrow \mathbb{R}^2(\mathbb{R})$ be a linear transformation defined by

$T(x, y, z) = (2x + y - z, 3x - 2y + 4z)$. Find the matrix of the linear transformation T with respect to the bases $B_1 = \{(1, 1, 1), (1, 1, 0), (1, 0, 0)\}$ and $B_2 = \{(1, 3), (1, 4)\}$.

Answer: $[T: B_1, B_2] = \begin{pmatrix} 3 & 11 & 5 \\ -1 & -8 & -3 \end{pmatrix}$.

8. Let $B = \{(2, 3), (3, 5)\}$ and $B^1 = \{(4, 3), (1, 2)\}$ be the bases of a vector space $\mathbb{R}^2(\mathbb{R})$. Find the transition matrix (or, change of basis matrix) from B to B^1 .

Answer: $P = \begin{pmatrix} 11 & -1 \\ -6 & 1 \end{pmatrix}$.